

## OSI Model, TCP/UDP, Packets, and Wireshark Intro

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 2 hours.

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### OSI Model

Video: [https://youtu.be/vv4y\\_uOneC0?si=J5EF\\_7rXlYrNlzlS](https://youtu.be/vv4y_uOneC0?si=J5EF_7rXlYrNlzlS)

1. Application Layer: used by network applications, ex: HTTP, HTTPS, FTP, SNMP, etc
2. Presentation Layer: receives data from Application Layer in the form of characters and numbers, which is converted to binary format
3. Session Layer: manages connection
4. Transport Layer: reliable transport (TCP/UDP)
5. Network Layer: IP routing
6. Data Link Layer: MAC address / local delivery
7. Physical Layer: electrical / cable / wifi signals

Each layer is a package of protocols.

Research Task: "What OSI layers do DNS, HTTPS, and IP operate on?"

DNS operates at the Application Layer (Layer 7).

HTTPS operates at the Application (Layer 7). [What layer is https? - SplicedOnline](#)  
However, it fully covers Layers 4-7. So: transport, session, and presentation as well.

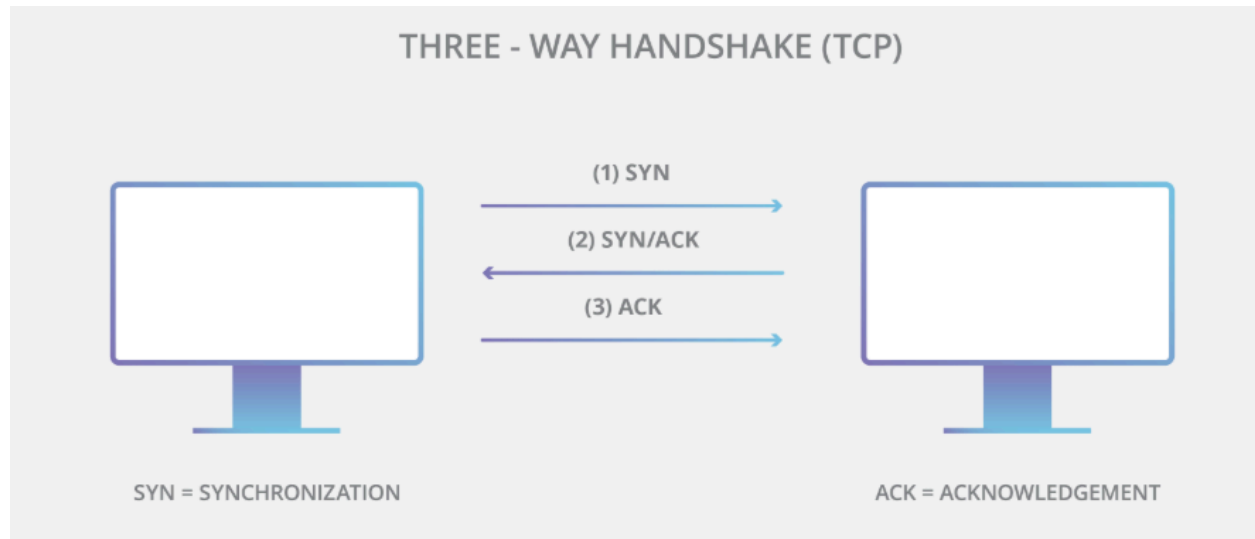
IP is Layer 3, the Network Layer.

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### TCP vs UDP + Packet Thinking

Article: [What is TCP/IP?](#) , [TCP vs. UDP - GeeksforGeeks](#)

IP (Internet Protocol) is the address system of the Internet and has the core function of delivering packets from source device to target device. However, IP does not handle packet ordering or error checking. This is often done by TCP, an acronym for Transmission Control Protocol. Note that IP is a connectionless protocol. Each unit of data is individually addressed and routed from the source device to the target device, and the target does not send an acknowledgment back to the source. TCP must be used in conjunction with IP in order to maintain a connection between the sender and the target to ensure packet order.



UDP stands for User Datagram Protocol. TCP is a reliable, connection-oriented transport protocol that ensures accurate and ordered data delivery. It is slower but more dependable. UDP is a fast, connectionless protocol that sends data without reliability guarantees. It is efficient for applications where speed is more important than accuracy.

| TCP                 | UDP                       |
|---------------------|---------------------------|
| Reliable            | Faster                    |
| Connection-oriented | Connectionless            |
| Ordered delivery    | No guarantee              |
| Used for HTTPS      | Used for gaming/streaming |

Why do games often use UDP? [Why Is Udp Used For Online Gaming | Robots.net](#)  
UDP has faster communication with low latency. A slight delay can result in a poor gaming experience. UDP's minimal overhead and lack of reliability mechanisms make it a suitable choice for real-time applications where low latency is crucial.

Why do websites use TCP?  
TCP is used because of its reliability, ordering, and error-handling capabilities. Reliable Data Transfer HTTP requires that the data sent from a server arrives intact and complete. TCP ensures this with connection-oriented three way handshake. No part of the web page is missing or corrupted during transmission.

Video: [How a packet moves through network - YouTube](#)

Packets contain: Source IP, destination IP, protocol, payload/data, and ports.

What information exists inside a packet?

Firstly, the header. The header contains essential metadata for routing and processing the packet. It typically includes:

- Source IP address
- Destination IP Address
- Protocol Information: transport layer protocol used (TCP/UDP)
- Sequence Number: Helps in reassembling packets in correct order at the destination
- Checksum: an error-checking code to ensure data integrity during transmission

Next, the payload. This is the actual data being transmitted. Ex: segment of a file, message, or any type of information sent over the network.

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## Wireshark Intro

[Wireshark • Go Deep | Download](#)

Tutorial Video: <https://youtu.be/TkCSr30UojM?si=FEbwcXzQFC6YO2n9>

Start capture. While capture runs: Open youtube, open edge, refresh discord, browse websites. Stop after 1-2 mins.

Try filters: dns, http, tcp, udp

Filter: ip.addr == YOUR\_IP

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## Mini Packet Analysis

Find a Client Hello inside TLS traffic.

What is a TLS Client Hello?

It is the initial message sent by a client during a TLS handshake. It serves as the starting point for establishing a secure communication channel between the client and the server.